

Home Network Security Lab

Building, Securing, and Monitoring a Virtualized Enterprise-Style Network

A hands-on SOC / Blue Team portfolio project covering network segmentation, firewall configuration, intrusion detection, SIEM deployment, and adversary simulation.

Component	Role
VMware Workstation Pro	Hypervisor hosting the entire lab
pfSense (Community Edition)	Firewall / router / DHCP server
Snort (pfSense package)	Network Intrusion Detection System (NIDS)
Wireshark	Manual packet-level traffic analysis
Wazuh (Manager + Agents)	SIEM — log correlation, FIM, alerting
Kali Linux	Attacker machine (Nmap, Hydra, Searchsploit)
Ubuntu Server	Linux victim / monitored endpoint
Windows 10	Windows victim / monitored endpoint (+ Sysmon)

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Project type: Self-directed SOC / Blue Team home lab

Scope: Phases A – E (network foundation through attack simulation)

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1. Executive Summary & Architecture

This project documents the design, build, and operation of a fully virtualized home SOC (Security Operations Center) lab. Using VMware Workstation Pro, a small isolated network was constructed behind a pfSense firewall, populated with an attacker machine (Kali Linux) and two victim/monitored endpoints (Ubuntu Server and Windows 10). Network-layer intrusion detection was provided by Snort, host-layer detection and log correlation by a Wazuh SIEM deployment, and manual packet inspection by Wireshark. The project then progressed into live attack simulation — brute-force credential attacks, vulnerability scanning, and exploit research — to validate that each detection layer worked as intended, and to practice the analyst workflow of investigating and explaining what was found.

Host System

16 GB RAM, 4 CPU cores, Linux host, VMware Workstation Pro as the hypervisor.

Lab Architecture

All virtual machines were placed on an isolated host-only network (VMnet2) with no direct route to the internet or the home LAN. pfSense bridged this internal segment (its LAN interface) to an internet-facing NAT network (VMnet8, its WAN interface), acting exactly as a home router would — providing DHCP, firewalling, and a controlled path out for updates, while keeping attack traffic fully contained.

Machine	Role	RAM	Disk	vCPU	IP Address
pfSense Firewall	Router / Firewall / IDS host	2 GB	20 GB	1	LAN: 192.168.1.1
Kali Linux	Attacker	2 GB	40 GB	2	192.168.1.100
Ubuntu Server	Victim / Wazuh Agent	2 GB	20 GB	1	192.168.1.102
Wazuh Manager	SIEM (Manager+Indexer +Dashboard)	4 GB	50 GB	2	192.168.1.103
Windows 10	Victim / Wazuh Agent + Sysmon	4 GB	60 GB	2	192.168.1.104

Detection stack summary: Snort inspects raw traffic crossing pfSense's LAN interface (network layer). Wazuh agents on Ubuntu Server and Windows report logs, file-integrity events, and rootcheck/anomaly data to the central Wazuh Manager (host layer). Wireshark was used for manual, on-demand packet inspection to verify and illustrate what Snort's signatures were matching.

2. Phase A — Hypervisor & Network Foundation

The first phase focused on installing VMware Workstation Pro and establishing an isolated internal network before any virtual machines were created — this ordering matters, since it guarantees every subsequent VM is placed behind pfSense from the moment it boots, rather than accidentally bridged onto the host's real network.

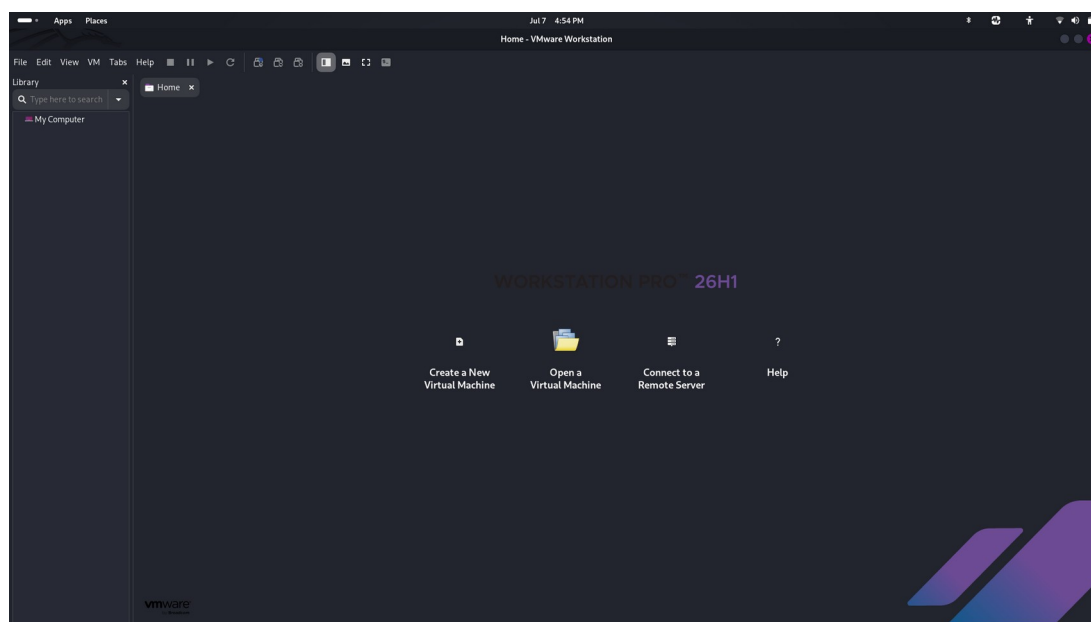


Figure 2.1 — VMware Workstation Pro freshly installed, before any virtual machines were created.

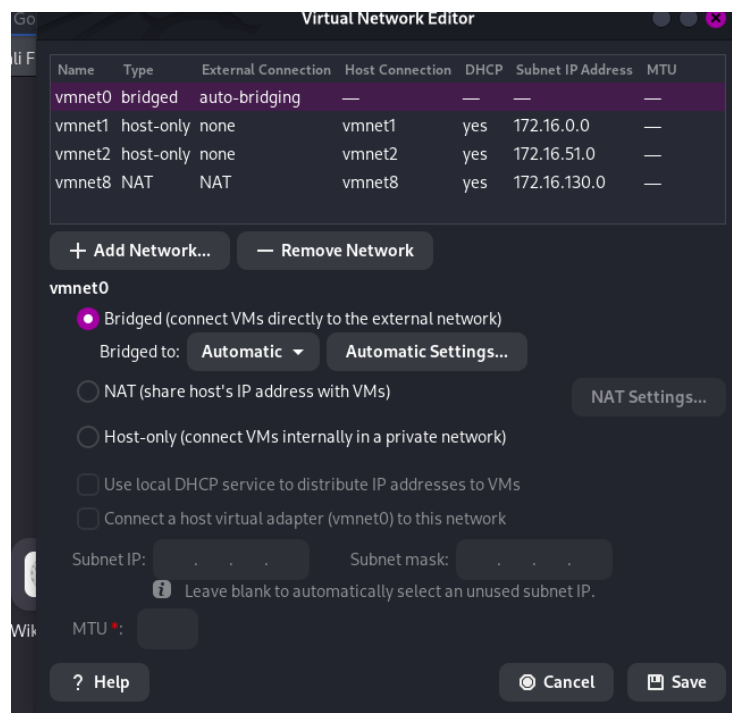


Figure 2.2 — Virtual Network Editor: VMnet2 configured as an isolated Host-only internal network, and VMnet8 retained as the NAT network used for pfSense's WAN/internet access.

pfSense Installation

pfSense Community Edition was installed as a VM with two network adapters: one on VMnet8 (NAT) acting as WAN, and one on VMnet2 (Host-only) acting as LAN. Its LAN interface was configured with the static address 192.168.1.1/24 and set up as the DHCP server for the entire internal segment.

⚠ **Issue encountered:** pfSense installer failed with a "Netgate Installer Error — the system is NOT installed" message.

✓ **Resolution:** The install log was inspected directly from the installer's root shell. Investigation traced the failure to two compounding factors: the VM's memory was set far too low (256 MB), and the WAN adapter could not reach pfSense's package repository during install. Increasing the VM's RAM to 2048 MB and confirming host/NAT internet connectivity resolved the install.

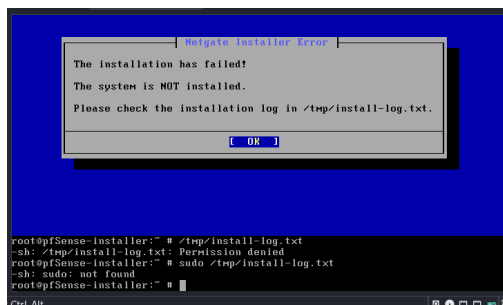


Figure 2.3 — pfSense installer failing with a generic install error.

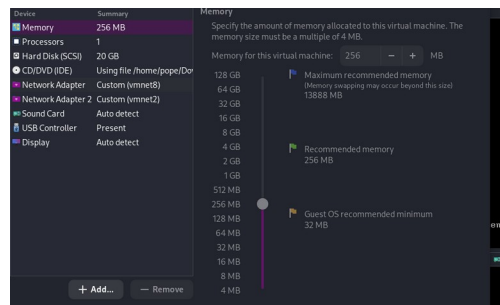


Figure 2.4 — Root cause #1: VM memory set to only 256 MB.

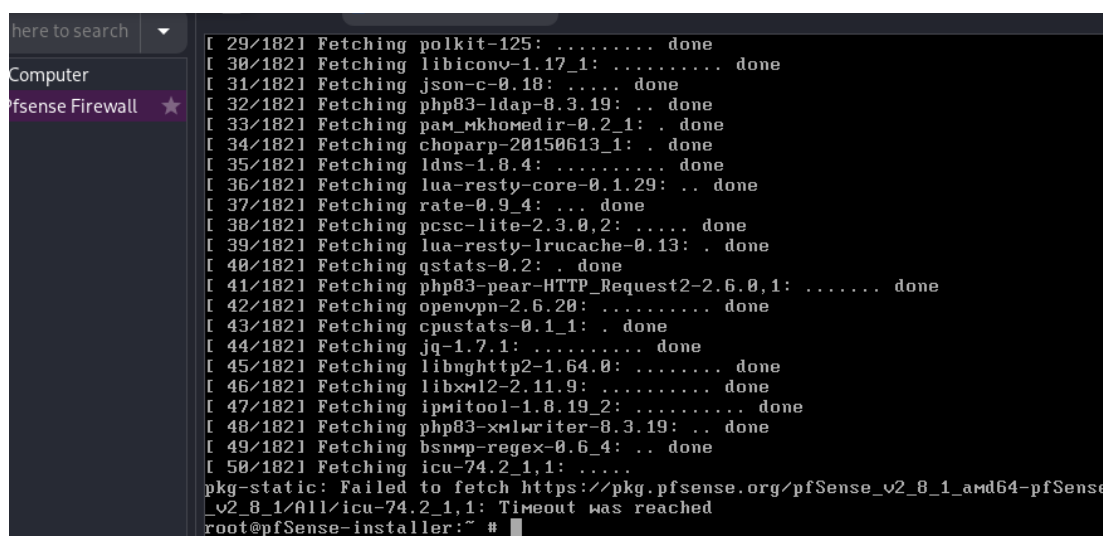


Figure 2.5 — Root cause #2: package fetch timing out, confirming a WAN/connectivity issue rather than a disk or media fault.

Result

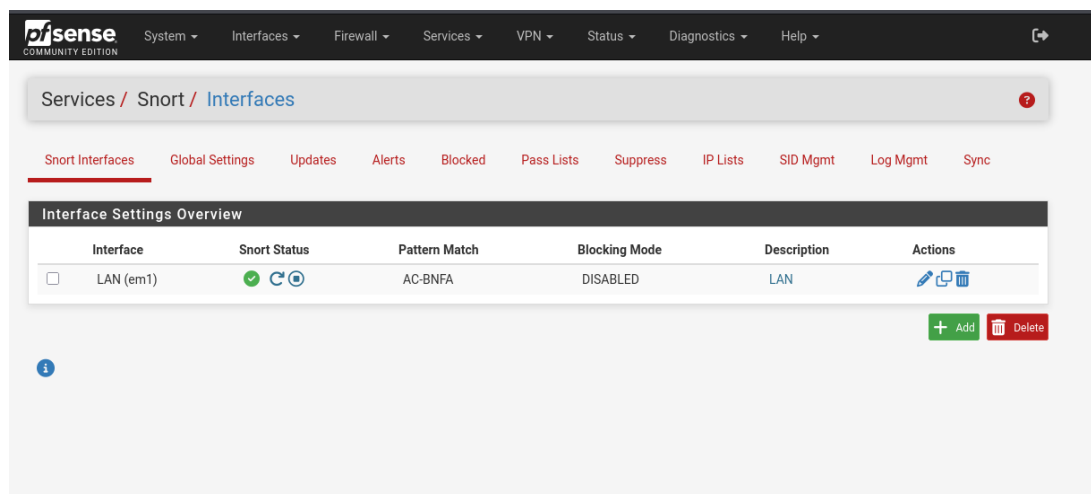


Figure 2.6 — pfSense Community Edition dashboard running successfully after the fixes above.

3. Phase B — Attacker & Victim Machines

With pfSense operational, the remaining lab machines were built: Kali Linux (attacker), Ubuntu Server (Linux victim/endpoint), and Windows 10 (Windows victim/endpoint). Every VM's network adapter was set to Custom → VMnet2, placing it on the same internal segment as pfSense's LAN interface so that all traffic is routed and monitored through the firewall.

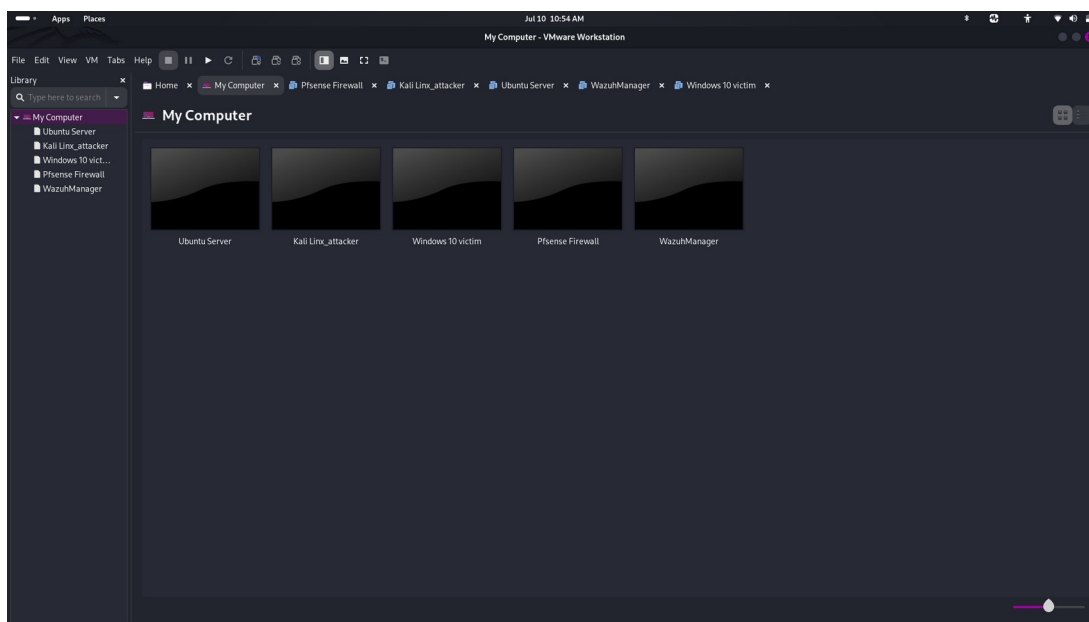


Figure 3.1 — All five virtual machines listed together in the VMware Workstation library.

⚠ Issue encountered: VMware's own local DHCP service was still enabled on VMnet2 at the same time pfSense was trying to serve DHCP on the same segment — a two-DHCP-server conflict causing intermittent wrong gateway/DNS settings.

✓ Resolution: "Use local DHCP service" was explicitly unchecked for VMnet2 in the Virtual Network Editor, leaving pfSense as the sole DHCP authority. Every VM then consistently received a clean 192.168.1.x lease directly from pfSense.

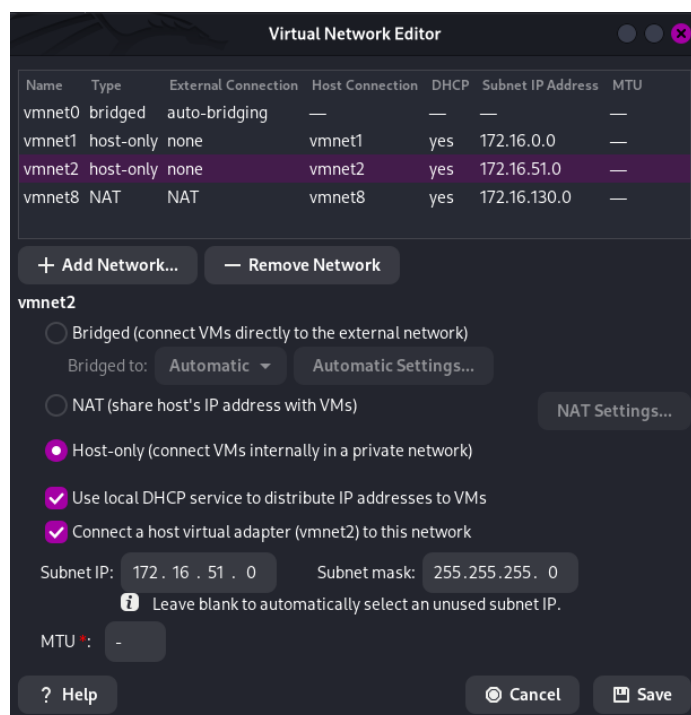


Figure 3.2 — Confirming DHCP leases correctly assigned from pfSense's 192.168.1.0/24 pool.

⚠ **Issue encountered:** Package installation on Ubuntu Server began failing with repeated "Temporary failure resolving 'archive.ubuntu.com'" errors.

✓ **Resolution:** Diagnosis isolated this to DNS resolution specifically (ICMP pings to 8.8.8.8 succeeded, confirming routing/NAT through pfSense worked) — resolved by confirming pfSense's DNS Resolver was actively running and correctly advertised via DHCP.

```

Error: Unable to fetch some archives, maybe run apt update or try with --fix-missing?
pope@popesays:~$ sudo apt update
Ign:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Ign:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Ign:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Err:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Temporary failure resolving 'security.ubuntu.com'
Ign:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Ign:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Err:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Temporary failure resolving 'archive.ubuntu.com'
Err:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Temporary failure resolving 'archive.ubuntu.com'
Err:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Temporary failure resolving 'archive.ubuntu.com'
All packages are up to date.
Warning: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/resolute/InRelease Temporary failure resolving 'archive.ubuntu.com'
Warning: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/resolute-updates/InRelease Temporary failure resolving 'archive.ubuntu.com'
Warning: Failed to fetch http://archive.ubuntu.com/ubuntu/dists/resolute-backports/InRelease Temporary failure resolving 'archive.ubuntu.com'
Warning: Failed to fetch http://security.ubuntu.com/ubuntu/dists/resolute-security/InRelease Temporary failure resolving 'security.ubuntu.com'
Warning: Some index files failed to download. They have been ignored, or old ones used instead.
pope@popesays:~$ sudo apt upgrade
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
pope@popesays:~$ sudo apt install net-tools
Installing:
net-tools
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
Download size: 151 kB
Space needed: 680 kB / 4,975 MB available
Ign:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 net-tools amd64 2.10-2ubuntu1
Ign:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 net-tools amd64 2.10-2ubuntu1
Ign:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 net-tools amd64 2.10-2ubuntu1
Err:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 net-tools amd64 2.10-2ubuntu1
Temporary failure resolving 'archive.ubuntu.com'
Error: Failed to fetch http://archive.ubuntu.com/ubuntu/pool/main/n/net-tools/net-tools_2.10-2ubuntu1_amd64.deb Temporary failure resolving 'archive.ubuntu.com'
Error: Unable to fetch some archives, maybe run apt update or try with --fix-missing?
pope@popesays:~$

```

Figure 3.3 — apt update failing with DNS resolution errors before the pfSense DNS configuration was corrected.

Network Discovery Verification

With all machines online, an Nmap ping sweep from Kali confirmed every host was reachable and correctly addressed on the shared internal segment before any security tooling was layered on top.

```

pope@popesays: ~
Session Actions Edit View Help

inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
roup default qlen 1000
    link/ether 00:0c:29:84:73:87 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.100/24 brd 192.168.1.255 scope global dynamic noprefixrou
e eth0
        valid_lft 5891sec preferred_lft 5891sec
        inet6 fe80::20c:29ff:fe84:7387/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(popec@popesays)~$ nmap -sn 192.168.1.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-09 10:48 -0400
Nmap scan report for pfSense.home.arpa (192.168.1.1)
Host is up (0.0015s latency).
MAC Address: 00:0C:29:6D:00:80 (VMware)
Nmap scan report for 192.168.1.63
Host is up (0.00041s latency).
MAC Address: 00:50:56:C0:00:02 (VMware)
Nmap scan report for 192.168.1.100
Host is up.
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.47 seconds
(popec@popesays)~$

```


Figure 3.4 — Nmap ping sweep discovering pfSense and all lab hosts on the internal network.

4. Phase C — Visibility: Snort IDS & Wireshark

This phase added network-layer detection. Snort was installed as a pfSense package and bound to the LAN interface so it could inspect all traffic crossing the internal segment. A mix of community rule categories (ET Open) and hand-written custom rules were enabled, and Wireshark was used alongside Snort to manually verify exactly what was on the wire during a simulated attack.

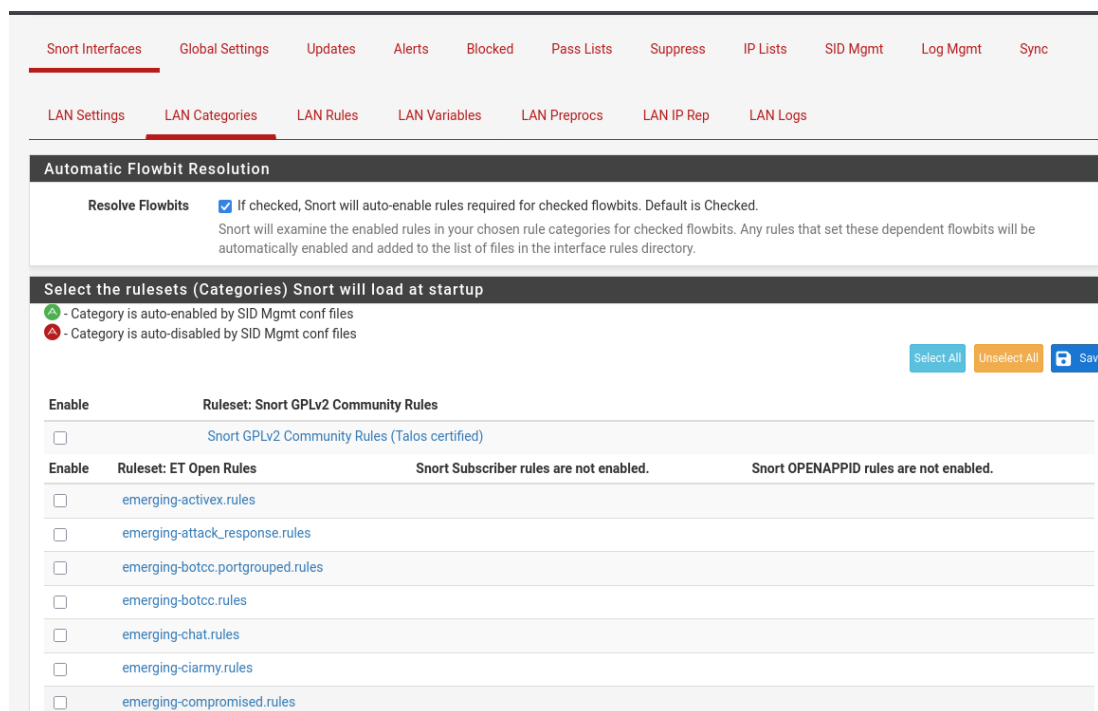


Figure 4.1 — Snort's rule Category management page on pfSense: ET Open rulesets (Scan, Exploit, Attack Response, Web Server/Client, Policy) enabled to cover reconnaissance, exploitation, and brute-force detection scenarios.

⚠ Issue encountered: Custom Snort rules failed to save, with the parser reporting "Fatal Error... Illegal direction specifier: \$HOME_NET".

✓ Resolution: The rule syntax was correct, but a corrupted arrow character introduced by copy-pasting from a rich-text source broke the parser. Rules were re-entered manually, one per line, using the exact ASCII arrow, which resolved the error.

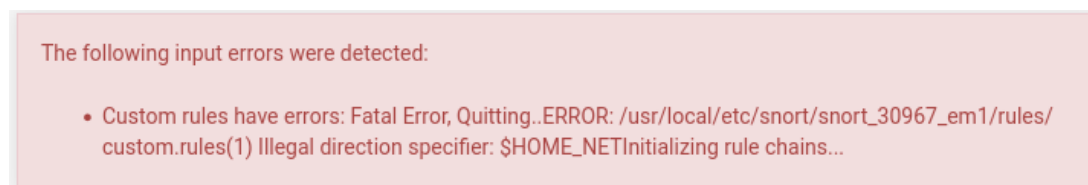


Figure 4.2 — Custom rule parsing error caused by a corrupted arrow character during copy-paste.

Detecting a Live Nmap Scan

With Snort running on the LAN interface, an Nmap SYN scan launched from Kali against Ubuntu Server was successfully flagged and logged as an alert.

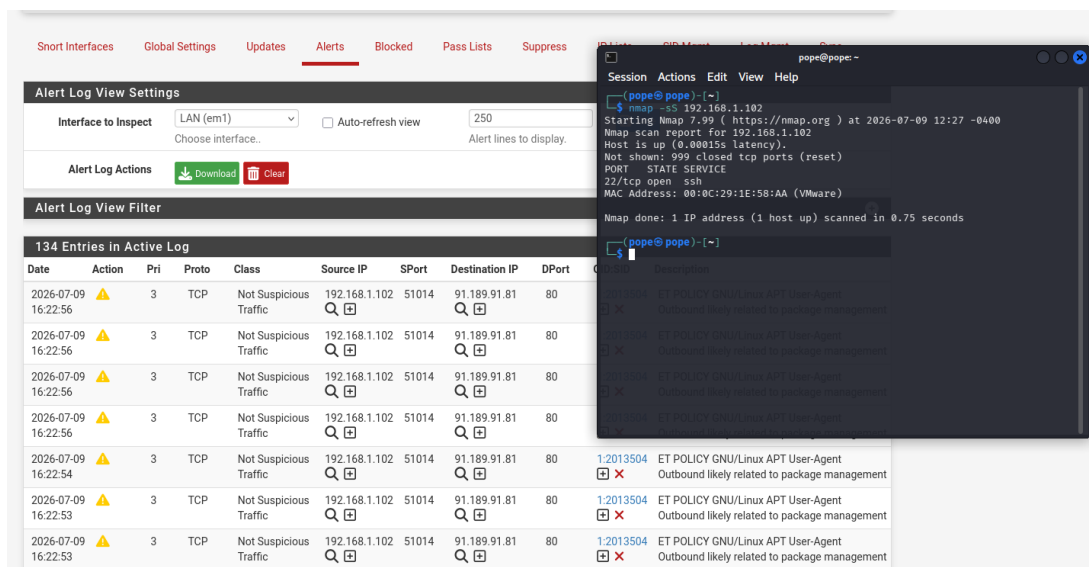


Figure 4.3 — A live Nmap SYN scan from Kali, with the corresponding Snort Alerts panel confirming detection on the LAN interface.

Issue encountered: Despite traffic clearly flowing between Kali and Ubuntu Server, Snort intermittently produced zero alerts, and a live tcpdump capture showed only broadcast/ARP traffic — not the actual unicast scan packets between the two hosts.

Resolution: VMware Workstation was silently blocking the LAN adapter's request for promiscuous mode — a hard hypervisor-level restriction, not just a guest-OS setting. Confirmed via tcpdump: cross-host traffic only became visible once promiscuous mode was actually granted at the host level. The fix required both a .vmx edit (`ethernet1.noPromisc = "FALSE"` and `isolation.tools.setPromiscuous.disable = "FALSE"`) and explicit device permissions on the host (`sudo chmod a+rw /dev/vmnet2`), followed by a full VM power-off/power-on cycle (an in-guest reboot alone does not reload .vmx changes).

```
h 100
16:13:49.837910 IP 192.168.1.102.44274 > 105.125.198.122.123: NTPv4, Client, len
gth 100
16:13:50.059910 IP 105.125.198.122.123 > 192.168.1.102.44274: NTPv4, Server, len
gth 100
16:13:51.365776 IP 192.168.1.102.52337 > 105.125.198.123.123: NTPv4, Client, len
gth 100
16:13:51.596101 IP 105.125.198.123.123 > 192.168.1.102.52337: NTPv4, Server, len
gth 100
16:13:51.879410 IP 192.168.1.102.49610 > 91.189.91.112.123: NTPv4, Client, lengt
h 100
16:13:52.180606 IP 91.189.91.112.123 > 192.168.1.102.49610: NTPv4, Server, lengt
h 100
16:13:53.686955 ARP, Request who-has 192.168.1.1 tell 192.168.1.102, length 46
16:13:53.687273 ARP, Reply 192.168.1.1 is-at 00:0c:29:64:00:80, length 28
76
13 packets captured
22 packets received by filter
0 packets dropped by kernel
12.0.1-BELKSH1root@pfSense:~$ tcpdump -l em1 -n host 192.168.1.1
02
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on em1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
16:15:00.009101 ARP, Request who-has 192.168.1.102 tell 192.168.1.100, length 46
```

Figure 4.4 — Before the fix: tcpdump shows only ARP/NTP traffic — cross-host scan traffic invisible.

```
20:50:31.560550 IP 192.168.1.100.61219 > 192.168.1.102.2393: tcp 0
20:50:31.560562 IP 192.168.1.100.61219 > 192.168.1.102.16113: tcp 0
20:50:31.560564 IP 192.168.1.102.427 > 192.168.1.100.61219: tcp 0
20:50:31.560567 IP 192.168.1.102.8402 > 192.168.1.100.61219: tcp 0
20:50:31.560570 IP 192.168.1.100.61219 > 192.168.1.102.6106: tcp 0
20:50:31.560572 IP 192.168.1.100.61219 > 192.168.1.102.1047: tcp 0
20:50:31.560574 IP 192.168.1.102.2393 > 192.168.1.100.61219: tcp 0
20:50:31.560576 IP 192.168.1.102.16113 > 192.168.1.100.61219: tcp 0
20:50:31.560579 IP 192.168.1.100.61219 > 192.168.1.102.7007: tcp 0
20:50:31.560883 IP 192.168.1.100.61219 > 192.168.1.102.49153: tcp 0
20:50:31.560889 IP 192.168.1.102.6106 > 192.168.1.100.61219: tcp 0
20:50:31.560892 IP 192.168.1.102.1047 > 192.168.1.100.61219: tcp 0
20:50:31.561820 IP 192.168.1.102.7007 > 192.168.1.100.61219: tcp 0
20:50:31.561828 IP 192.168.1.102.49153 > 192.168.1.100.61219: tcp 0
20:50:31.561831 IP 192.168.1.100.61219 > 192.168.1.102.65129: tcp 0
20:50:31.561833 IP 192.168.1.100.61219 > 192.168.1.102.548: tcp 0
20:50:31.561836 IP 192.168.1.100.61219 > 192.168.1.102.10609: tcp 0
20:50:31.561839 IP 192.168.1.100.61219 > 192.168.1.102.40155: tcp 0
20:50:31.561841 IP 192.168.1.100.61219 > 192.168.1.102.3052: tcp 0
```

Figure 4.5 — After the fix: tcpdump captures full Kali → Ubuntu port-scan traffic, confirming promiscuous mode active.

Manual Packet Verification with Wireshark

To visually confirm what Snort's signatures were matching, pfSense's built-in Diagnostics → Packet Capture tool was used to capture LAN traffic during a fresh scan, and the resulting .pcap was analyzed in Wireshark using the display filter `tcp.flags.syn==1 and tcp.flags.ack==0`, which isolates bare SYN packets — the signature of a SYN/stealth port scan.

```

21:04:45.381335 IP 192.168.1.102.17877 > 192.168.1.100.38674: tcp 0
21:04:45.381480 IP 192.168.1.102.3784 > 192.168.1.100.38674: tcp 0
21:04:45.381484 IP 192.168.1.102.1038 > 192.168.1.100.38674: tcp 0
21:04:45.381683 IP 192.168.1.102.26214 > 192.168.1.100.38674: tcp 0
21:04:45.381687 IP 192.168.1.102.1277 > 192.168.1.100.38674: tcp 0
21:04:45.381688 IP 192.168.1.100.38674 > 192.168.1.102.3000: tcp 0
21:04:45.381690 IP 192.168.1.100.38674 > 192.168.1.102.1084: tcp 0
21:04:45.381692 IP 192.168.1.102.5962 > 192.168.1.100.38674: tcp 0
21:04:45.381694 IP 192.168.1.102.1106 > 192.168.1.100.38674: tcp 0
21:04:45.382437 IP 192.168.1.102.1147 > 192.168.1.100.38674: tcp 0
21:04:45.382441 IP 192.168.1.102.1048 > 192.168.1.100.38674: tcp 0
21:04:45.382444 IP 192.168.1.102.3031 > 192.168.1.100.38674: tcp 0
21:04:45.382446 IP 192.168.1.102.1072 > 192.168.1.100.38674: tcp 0
21:04:45.382448 IP 192.168.1.100.38674 > 192.168.1.102.34571: tcp 0
21:04:45.382450 IP 192.168.1.100.38674 > 192.168.1.102.3914: tcp 0
21:04:45.382452 IP 192.168.1.102.3000 > 192.168.1.100.38674: tcp 0
21:04:45.382453 IP 192.168.1.102.1084 > 192.168.1.100.38674: tcp 0
21:04:45.382455 IP 192.168.1.100.38674 > 192.168.1.102.8042: tcp 0
21:04:45.382457 IP 192.168.1.100.38674 > 192.168.1.102.27355: tcp 0

```

```

Nmap done: 1 IP address (1 host up) scanned in 0.94 seconds

(pope@pope)-[~]
$ sudo nmap -sS 192.168.1.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-10 17:04 -0400
Nmap scan report for 192.168.1.102
Host is up (0.0016s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
MAC Address: 00:0C:29:1E:58:AA (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.01 seconds

(pope@pope)-[~]
$

```

Figure 4.6 — pfSense's Packet Capture tool configured to capture LAN traffic during a scan.

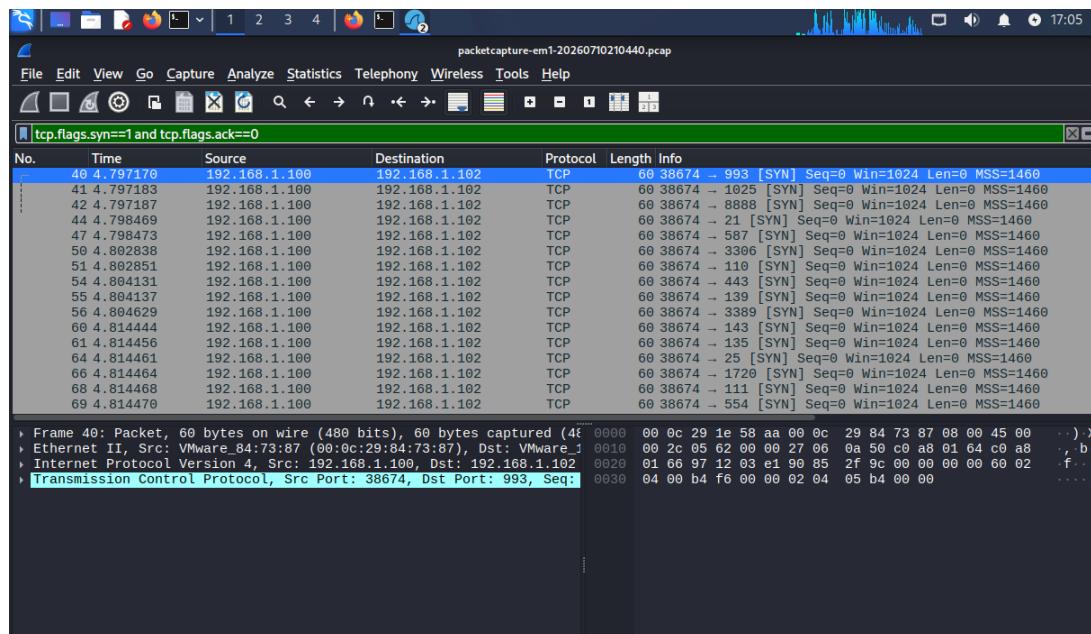


Figure 4.7 — Wireshark view filtered to SYN-only packets: dozens of SYN packets from Kali hitting many destination ports on Ubuntu Server in rapid succession — the fingerprint of a port scan.

5. Phase D — Wazuh SIEM Deployment

This phase added the host-based / SIEM layer. A Wazuh Manager (bundling the Manager, Indexer, and Dashboard components) was deployed as its own VM. Wazuh agents were then installed on Ubuntu Server and Windows 10 so that host-level telemetry — logs, file integrity events, rootcheck anomaly scans — would flow into a single, searchable dashboard.

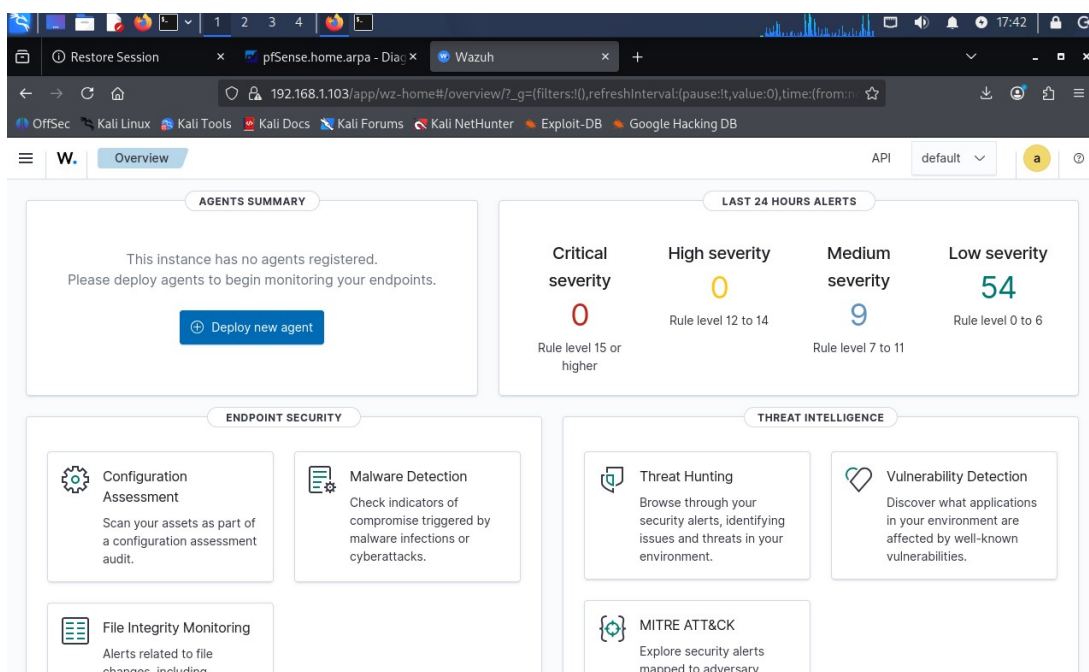


Figure 5.1 — The Wazuh dashboard Overview page immediately after deployment, before any agents were enrolled.

⚠ Issue encountered: Installing the Wazuh agent via a manually-typed wget URL for the .deb package consistently returned 403 Forbidden.

✓ Resolution: The direct file URL was stale. Switching to the Wazuh dashboard's built-in Deploy New Agent wizard generated a correct, version-matched install command automatically, which resolved the download immediately.

```
2026-07-10 22:02:45 - https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.6-1_am64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 108.157.78.119, 108.157.78.23, 108.157.78.21, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|108.157.78.119|:443... connected.
HTTP request sent, awaiting response... 403 Forbidden
2026-07-10 22:02:47 ERROR 403: Forbidden.

pope@popesays:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.6-1_am64.deb && sudo WAZUH_MANAGER='192.168.1.102'
--2026-07-10 22:03:06-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.6-1_am64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 108.157.78.21, 108.157.78.119, 108.157.78.23, 108.157.78.113, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|108.157.78.21|:443... connected.
HTTP request sent, awaiting response... 403 Forbidden
2026-07-10 22:03:37 ERROR 403: Forbidden.

pope@popesays:~$ ip a
1: lo: LOOPBACK, UP, LOWER_UP, mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:1e:58:aa brd ff:ff:ff:ff:ff:ff
    altname ens33
    altname enx00c291e58aa
    inet 192.168.1.102/24 metric 100 brd 192.168.1.255 scope global dynamic ens33
        valid_lft 6303sec preferred_lft 6303sec
    inet6 fe80::20c:29ff:fe1e:58aa/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever

pope@popesays:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.6-1_am64.deb && sudo WAZUH_MANAGER='192.168.1.102' WAZUH_AGENT
NAME='UbuntuServer' dkg -i ./wazuh-agent_4.14.6-1_am64.deb
--2026-07-10 22:05:00-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.6-1_am64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 108.157.78.119, 108.157.78.23, 108.157.78.113, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|108.157.78.119|:443... connected.
HTTP request sent, awaiting response... 403 Forbidden
2026-07-10 22:05:32 ERROR 403: Forbidden.

pope@popesays:~$ ^V

^C
command not found
pope@popesays:~$
pope@popesays:~$
pope@popesays:~$ wget https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.6-1_am64.deb && sudo WAZUH_MANAGER='192.168.1.102' WAZUH_AGENT
NAME='UbuntuServer' dkg -i ./wazuh-agent_4.14.6-1_am64.deb
--2026-07-10 22:09:18-- https://packages.wazuh.com/4.x/apt/pool/main/w/wazuh-agent/wazuh-agent_4.14.6-1_am64.deb
Resolving packages.wazuh.com (packages.wazuh.com)... 108.157.78.119, 108.157.78.23, 108.157.78.113, ...
Connecting to packages.wazuh.com (packages.wazuh.com)|108.157.78.119|:443... connected.
HTTP request sent, awaiting response... 403 Forbidden
2026-07-10 22:09:21 ERROR 403: Forbidden.
```

Figure 5.2 — Repeated 403 Forbidden errors from a manually-constructed package URL.

⚠ **Issue encountered:** After installation, the Ubuntu Server agent could not enroll: ossec.log showed repeated errors — "Unable to connect to enrollment service at [192.168.1.102]:1515".

✓ **Resolution:** The agent was misconfigured to contact itself (192.168.1.102) instead of the actual Wazuh Manager (192.168.1.103). Correcting the address field inside ossec.conf and restarting the agent service resolved the enrollment failure.

```
pope@popesays:~$ cat /var/ossec/logs/ossec.log | tail -30
cat: /var/ossec/logs/ossec.log: Permission denied
pope@popesays:~$ sudo cat /var/ossec/logs/ossec.log | tail -30
2026/07/11 12:21:33 wazuh-agentd: INFO: Requesting a key from server: 192.168.1.102
2026/07/11 12:21:33 wazuh-agentd: ERROR: (1208): Unable to connect to enrollment service at '[192.168.1.102]:1515'
2026/07/11 12:22:08 wazuh-agentd: INFO: Requesting a key from server: 192.168.1.102
2026/07/11 12:22:08 wazuh-agentd: ERROR: (1208): Unable to connect to enrollment service at '[192.168.1.102]:1515'
2026/07/11 12:22:48 wazuh-agentd: INFO: Requesting a key from server: 192.168.1.102
2026/07/11 12:22:48 wazuh-agentd: ERROR: (1208): Unable to connect to enrollment service at '[192.168.1.102]:1515'
2026/07/11 12:23:33 wazuh-agentd: INFO: Requesting a key from server: 192.168.1.102
2026/07/11 12:23:33 wazuh-agentd: ERROR: (1208): Unable to connect to enrollment service at '[192.168.1.102]:1515'
2026/07/11 12:24:23 wazuh-agentd: INFO: Requesting a key from server: 192.168.1.102
2026/07/11 12:24:23 wazuh-agentd: ERROR: (1208): Unable to connect to enrollment service at '[192.168.1.102]:1515'
2026/07/11 12:25:18 wazuh-agentd: INFO: Requesting a key from server: 192.168.1.102
2026/07/11 12:25:18 wazuh-agentd: ERROR: (1208): Unable to connect to enrollment service at '[192.168.1.102]:1515'
2026/07/11 12:26:18 wazuh-agentd: INFO: Requesting a key from server: 192.168.1.102
2026/07/11 12:26:18 wazuh-agentd: ERROR: (1208): Unable to connect to enrollment service at '[192.168.1.102]:1515'
2026/07/11 12:27:18 wazuh-agentd: INFO: Requesting a key from server: 192.168.1.102
2026/07/11 12:27:18 wazuh-agentd: ERROR: (1208): Unable to connect to enrollment service at '[192.168.1.102]:1515'
2026/07/11 12:28:18 wazuh-agentd: INFO: Requesting a key from server: 192.168.1.102
2026/07/11 12:28:18 wazuh-agentd: ERROR: (1208): Unable to connect to enrollment service at '[192.168.1.102]:1515'
2026/07/11 12:29:18 wazuh-agentd: INFO: Requesting a key from server: 192.168.1.102
2026/07/11 12:29:18 wazuh-agentd: ERROR: (1208): Unable to connect to enrollment service at '[192.168.1.102]:1515'
```

Figure 5.3 — ossec.log revealing the agent was attempting to enroll against its own IP address instead of the Wazuh Manager's.

Both Agents Active

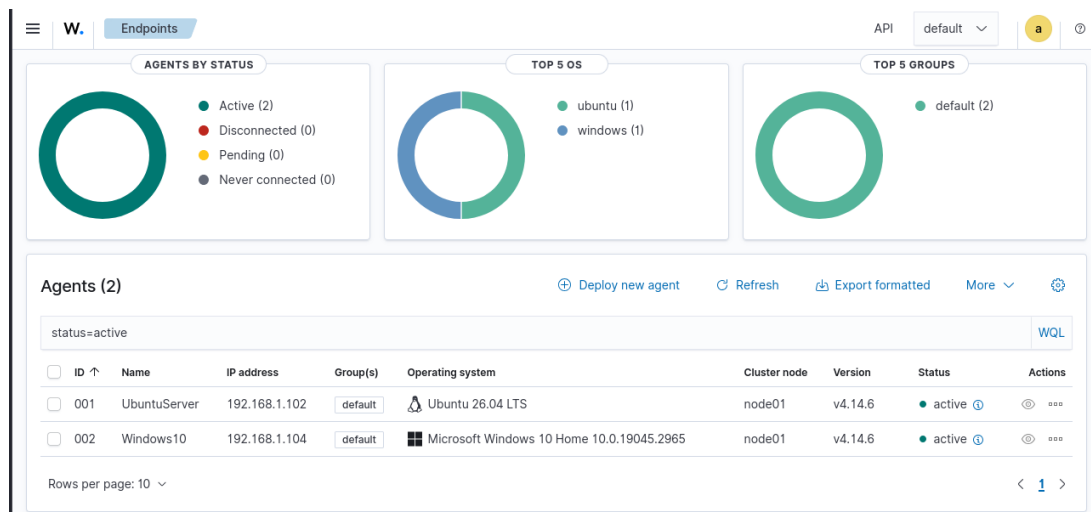


Figure 5.4 — Both endpoints (UbuntuServer and Windows10) showing as Active in the Wazuh Endpoints view.

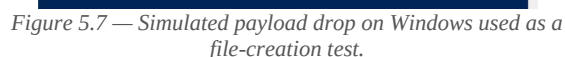
Sysmon Deployment on Windows

Sysmon (Microsoft Sysinternals), configured with the community-standard SwiftOnSecurity ruleset, was installed on the Windows 10 endpoint to generate richer process, network, and file-level telemetry than default Windows Event Logs alone.

⚠ **Issue encountered:** The Sysmon install command failed with "'.\Sysmon64.exe' is not recognized".

Figure 5.5 — Sysmon install command failing due to running it from the wrong working directory.

FIM was configured with realtime monitoring on Ubuntu Server's /etc directory and tested by creating and deleting a marker file. On Windows, a directory not covered by the default FIM scope (C:\Windows\Temp) was added explicitly, and a test file drop was used to simulate a dropped payload.

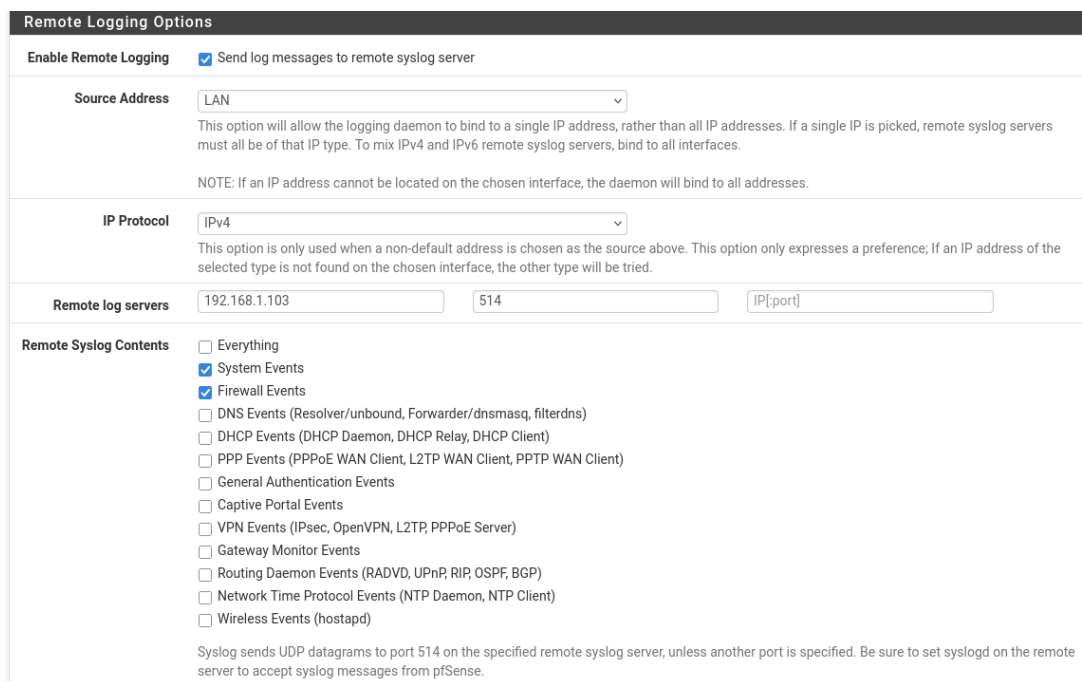


Forwarding pfSense Logs to Wazuh via Syslog

pfSense's Remote Logging feature was configured to forward System and Firewall events to the Wazuh Manager over UDP 514, with a matching <remote> listener block added to the Wazuh Manager's own ossec.conf.

⚠ Issue encountered: The Remote Logging form rejected saving with "A valid IP address/hostname or IP/hostname:port must be specified for remote syslog server #2".

✓ Resolution: The IP and port had been entered into two separate server slots rather than combined into a single IP:port value in one field. Combining them as 192.168.1.103:514 in the first slot resolved the validation error.



Remote Logging Options

Enable Remote Logging ☒ Send log messages to remote syslog server

Source Address LAN
This option will allow the logging daemon to bind to a single IP address, rather than all IP addresses. If a single IP is picked, remote syslog servers must all be of that IP type. To mix IPv4 and IPv6 remote syslog servers, bind to all interfaces.
NOTE: If an IP address cannot be located on the chosen interface, the daemon will bind to all addresses.

IP Protocol IPv4
This option is only used when a non-default address is chosen as the source above. This option only expresses a preference; If an IP address of the selected type is not found on the chosen interface, the other type will be tried.

Remote log servers 192.168.1.103 514 IP[:port]

Remote Syslog Contents

- ☐ Everything
- ☒ System Events
- ☒ Firewall Events
- ☐ DNS Events (Resolver/unbound, Forwarder/dnsmasq, filterdns)
- ☐ DHCP Events (DHCP Daemon, DHCP Relay, DHCP Client)
- ☐ PPP Events (PPPoE WAN Client, L2TP WAN Client, PPTP WAN Client)
- ☐ General Authentication Events
- ☐ Captive Portal Events
- ☐ VPN Events (IPsec, OpenVPN, L2TP, PPPoE Server)
- ☐ Gateway Monitor Events
- ☐ Routing Daemon Events (RADVD, UPnP, RIP, OSPF, BGP)
- ☐ Network Time Protocol Events (NTP Daemon, NTP Client)
- ☐ Wireless Events (hostapd)

Syslog sends UDP datagrams to port 514 on the specified remote syslog server, unless another port is specified. Be sure to set syslogd on the remote server to accept syslog messages from pfSense.

Figure 5.9 — pfSense's Remote Logging Options correctly configured: forwarding to 192.168.1.103:514 over UDP.

End-to-end delivery was confirmed by grepping the Wazuh Manager's archive log for pfSense's IP address after enabling full event archiving, which surfaced genuine pfSense system events that had successfully traversed the syslog pipeline.


```

2026 Jul 11 15:12:23 (UbuntuServer) amy->syscollector {'data': {'checksum': 'aa3e4737c4b7659a08371ed4b6d42a5f010ff95', 'inode': 170513, 'item_id': 'b135d07d4226d0e0f590e9d98b2cafa9ad7927e', 'local_ip': '192.168.1.102', 'local_port': 41439, 'pid': 3147, 'process': 'wazuh-agent', 'protocol': 'tcp', 'remote_ip': '192.168.1.103', 'remote_port': 1514, 'rx_queue': 0, 'scan_time': '2026-07-11 15:12:21', 'state': 'established', 'tx_queue': 0}, 'operation': 'DELETED', 'type': 'dbsync_ports'}
2026 Jul 11 15:13:33 wazuh-server->journald Jul 11 15:13:32 wazuh-server opensearch-dashboards[2951]: {"type": "response", "timestamp": "2026-07-11T15:13:32Z", "tags": [], "pid": 2951, "method": "get", "statusCode": 200, "req": {"url": "/api/saved_objects/_find?default_search_operator=AND&has_reference=7B22type:223a22visualizations/222c222d223a22events_histogram/2227D&page=1&per_page=100&search_fields=title:5E3&search_fields=description&type=augment-vis", "method": "get", "headers": {"host": "192.168.1.103", "user-agent": "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0", "accept": "*/", "accept-language": "en-US,en;q=0.5", "accept-encoding": "gzip, deflate, br, zstd", "referrer": "https://192.168.1.103/app/threat-hunting", "content-type": "application/json", "osd-version": "2.19.5", "osd-xsrf": "osd-fetch", "connection": "keep-alive", "sec-fetch-dest": "empty", "sec-fetch-mode": "cors", "sec-fetch-site": "same-origin", "priority": "u=0", "remoteaddress": "192.168.1.100", "useragent": "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0", "referrer": "https://192.168.1.103/app/threat-hunting"}, "res": {"statusCode": 200, "responseTime": 244, "contentType": "9", "message": "GET /api/saved_objects/_find?default_search_operator=AND&has_reference=7B22type:223a22visualizations/222c222d223a22events_histogram/2227D&page=1&per_page=100&search_fields=title:5E3&search_fields=description&type=augment-vis 200 244ms - 9.0B"}
2026 Jul 11 15:13:35 wazuh-server->journald Jul 11 15:13:33 wazuh-server opensearch-dashboards[2951]: {"type": "response", "timestamp": "2026-07-11T15:13:32Z", "tags": [], "pid": 2951, "method": "post", "statusCode": 200, "req": {"url": "/internal/search/opensearch", "method": "post", "headers": {"host": "192.168.1.103", "user-agent": "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0", "accept": "*/", "accept-language": "en-US,en;q=0.5", "accept-encoding": "gzip, deflate, br, zstd", "referrer": "https://192.168.1.103/app/threat-hunting", "content-type": "application/json", "osd-version": "2.19.5", "osd-xsrf": "osd-fetch", "content-length": "2502", "origin": "https://192.168.1.103", "connection": "keep-alive", "sec-fetch-dest": "empty", "sec-fetch-mode": "cors", "sec-fetch-site": "same-origin", "priority": "u=4", "remoteaddress": "192.168.1.100", "useragent": "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0", "referrer": "https://192.168.1.103/app/threat-hunting"}, "res": {"statusCode": 200, "responseTime": 596, "contentType": "9", "message": "POST /internal/search/opensearch 200 596ms - 9.0B"}
2026 Jul 11 15:14:23 wazuh-server->journald Jul 11 15:14:23 dhp6c[10512]: Sending Solicit
2026 Jul 11 15:14:49 wazuh-server opensearch-dashboards[2951]: {"type": "response", "timestamp": "2026-07-11T15:14:48Z", "tags": [], "pid": 2951, "method": "get", "statusCode": 200, "req": {"url": "/api/saved_objects/_find?default_search_operator=AND&has_reference=7B22type:223a22visualizations/222c222d223a22events_histogram/2227D&page=1&per_page=100&search_fields=title:5E3&search_fields=description&type=augment-vis", "method": "get", "headers": {"host": "192.168.1.103", "user-agent": "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0", "accept": "*/", "accept-language": "en-US,en;q=0.5", "accept-encoding": "gzip, deflate, br, zstd", "referrer": "https://192.168.1.103/app/threat-hunting", "content-type": "application/json", "osd-version": "2.19.5", "osd-xsrf": "osd-fetch", "content-length": "2502", "origin": "https://192.168.1.103", "connection": "keep-alive", "sec-fetch-dest": "empty", "sec-fetch-mode": "cors", "sec-fetch-site": "same-origin", "priority": "u=4", "remoteaddress": "192.168.1.100", "useragent": "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0", "referrer": "https://192.168.1.103/app/threat-hunting"}, "res": {"statusCode": 200, "responseTime": 104, "contentType": "9", "message": "POST /internal/search/opensearch 200 104ms - 9.0B"}
2026 Jul 11 15:14:51 wazuh-server->journald Jul 11 15:14:49 wazuh-server opensearch-dashboards[2951]: {"type": "response", "timestamp": "2026-07-11T15:14:48Z", "tags": [], "pid": 2951, "method": "post", "statusCode": 200, "req": {"url": "/internal/search/opensearch", "method": "post", "headers": {"host": "192.168.1.103", "user-agent": "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0", "accept": "*/", "accept-language": "en-US,en;q=0.5", "accept-encoding": "gzip, deflate, br, zstd", "referrer": "https://192.168.1.103/app/threat-hunting", "content-type": "application/json", "osd-version": "2.19.5", "osd-xsrf": "osd-fetch", "content-length": "2502", "origin": "https://192.168.1.103", "connection": "keep-alive", "sec-fetch-dest": "empty", "sec-fetch-mode": "cors", "sec-fetch-site": "same-origin", "priority": "u=4", "remoteaddress": "192.168.1.100", "useragent": "Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0", "referrer": "https://192.168.1.103/app/threat-hunting"}, "res": {"statusCode": 200, "responseTime": 777, "contentType": "9", "message": "POST /internal/search/opensearch 200 777ms - 9.0B"}
2026 Jul 11 15:16:14 wazuh-server->journald Jul 11 15:16:14 dhp6c[10512]: Sending Solicit
2026 Jul 11 15:16:47 wazuh-server sudo[13941]: wazuh-user : TTY=ttty1 ; PWD=/home/wazuh-user : USER=root ; COMMAND=/usr/bin/grep 192.168.1.1 /var/ossec/logs/archives/archives.log
2026 Jul 11 15:17:05 wazuh-server->journald Jul 11 15:17:06 kernel: em0: link state changed to DOWN
2026 Jul 11 15:17:06 wazuh-server->journald Jul 11 15:17:06 check_reload_status[5051]: Linkup starting em0
2026 Jul 11 15:17:07 wazuh-server->journald Jul 11 15:17:06 dpinger[713341]: WAN_DHCP 172.16.130.2: sendto error: 50
2026 Jul 11 15:17:07 wazuh-server->journald Jul 11 15:17:07 wazuh-server: }$

```

Figure 5.10 — Wazuh Manager's archive log confirming pfSense system/kernel events successfully arriving via the syslog forwarder.

Known limitation: pfSense's Snort package logs its own intrusion alerts to a local file rather than through pfSense's general remote-syslog forwarder, so Snort-specific alerts do not currently appear inside the Wazuh dashboard — they remain visible locally on pfSense's own Alerts page. This is a documented architectural gap rather than a misconfiguration.

Investigating an Unexplained Alert Burst

During routine review, the Threat Hunting view showed dozens of "Host-based anomaly detection event (rootcheck)" alerts firing within seconds of one another, which initially appeared to be a runaway loop or misconfiguration.

timestamp	agent.name	rule.description	rule.level	rule.id
Jul 13, 2026 @ 05:48:20.297	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:19.589	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:17.812	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:16.982	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:16.267	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:15.626	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:14.799	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:14.271	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:13.628	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:12.096	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:10.684	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:09.371	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:08.002	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:06.669	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510
Jul 13, 2026 @ 05:48:04.355	UbuntuServer	Host-based anomaly detection event (rootcheck).	7	510

Figure 5.11 — A rapid burst of rule 510 ("rootcheck") alerts from UbuntuServer, initially flagged for investigation.

Issue encountered: Whether the burst of rootcheck alerts represented a real problem or was caused by the recently-enabled remote syslog forwarding.

Resolution: Cross-referencing timestamps in ossec.log for "Starting rootcheck scan" showed only five scan-start events spread across three days — each coinciding with a deliberate agent/manager restart performed during earlier configuration changes. Since scan-on-start is enabled by default, each restart triggers one full rootcheck pass, producing many alerts within a short window before going quiet again. Manager resource checks also showed a stable, healthy system. Conclusion: expected, one-time behavior tied to configuration restarts — unrelated to syslog forwarding.

```

Jul 13 10:22:10 wazuh-server eno[3745]: Started wazuh-analysisd...
Jul 13 10:22:12 wazuh-server eno[3745]: Started wazuh-syscheckd...
Jul 13 10:22:13 wazuh-server eno[3745]: Started wazuh-remoted...
Jul 13 10:22:14 wazuh-server eno[3745]: Started wazuh-logcollector...
Jul 13 10:22:15 wazuh-server eno[3745]: Started wazuh-monitord...
Jul 13 10:22:15 wazuh-server eno[4233]: 2026-07-13 09:22:15 wazuh-modulesd:router: INFO: I
Jul 13 10:22:15 wazuh-server eno[4233]: 2026-07-13 09:22:15 wazuh-modulesd:content_manager
Jul 13 10:22:15 wazuh-server eno[4233]: 2026-07-13 09:22:15 wazuh-modulesd:inventory-harve
Jul 13 10:22:16 wazuh-server eno[3745]: Started wazuh-modulesd...
Jul 13 10:22:16 wazuh-server eno[3745]: Completed.
wazuh-user@wazuh-server ~$ free -h
             total        used        free      shared  buff/cache   available
Mem:          3.8Gi          3.1Gi          120Mi          11Mi          588Mi          466Mi
Swap:          0B           0B           0B
wazuh-user@wazuh-server ~$ top -bnl 1 head -20
top - 11:42:33 up 1:21, 1 user, load average: 0.32, 0.35, 0.26
tasks: 166 total, 1 running, 159 sleeping, 0 stopped, 0 zombie
%mems: 5.1 us, 12.9 sy, 0.0 ni, 82.1 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3919.0 total, 160.4 free, 3171.3 used, 587.3 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 499.5 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 7437 wazuh-u*   20   0 224956   3320  2852 R   16.7   0.1   0:00.05 top
 4125 wazuh    20   0 1163920 13660  6224 S   11.1   0.3   0:35.77 wazuh-remoted
    1 root      20   0 106692   1360   768 S    0.0   0.3   0:04.31 systemd
    2 root      20   0      0      0      0 S    0.0   0.0   0:00.02 kthreadd
    3 root      0 -20     0      0      0 I    0.0   0.0   0:00.00 rcu_gp
    4 root      0 -20     0      0      0 I    0.0   0.0   0:00.00 rcu_par_gp
    5 root      0 -20     0      0      0 I    0.0   0.0   0:00.00 slab_flushuq
    6 root      0 -20     0      0      0 I    0.0   0.0   0:00.00 netns
   10 root      0 -20     0      0      0 I    0.0   0.0   0:00.00 mnt_percpu_wq
   11 root      20   0      0      0      0 I    0.0   0.0   0:00.00 rcu_tasks_kthread
   12 root      20   0      0      0      0 I    0.0   0.0   0:00.00 rcu_tasks_rude_kth
   13 root      20   0      0      0      0 I    0.0   0.0   0:00.00 rcu_tasks_trace_kth
   14 root      20   0      0      0      0 S    0.0   0.0   0:01.11 ksoftirqd/0
wazuh-user@wazuh-server ~$
wazuh-user@wazuh-server ~$

```

Figure 5.12 — Confirming Wazuh Manager stability (RAM/CPU load healthy) while investigating the alert burst.

```

wazuh-user@wazuh-server ~$ sudo grep -i "Starting rootcheck" /var/ossec/logs/ossec.log
2026-07-10 21:30:56 rootcheck: INFO: Starting rootcheck scan.
2026-07-11 11:57:13 rootcheck: INFO: Starting rootcheck scan.
2026-07-11 14:40:17 rootcheck: INFO: Starting rootcheck scan.
2026-07-11 15:11:44 rootcheck: INFO: Starting rootcheck scan.
2026-07-13 09:22:11 rootcheck: INFO: Starting rootcheck scan.
wazuh-user@wazuh-server ~$
wazuh-user@wazuh-server ~$

```

Figure 5.13 — ossec.log showing only five rootcheck scan-start events across three days, each aligned with a known restart.

6. Phase E — Attack Simulation & Detection Exercises

With detection tooling fully wired up across both the network layer (Snort) and host layer (Wazuh), Phase E ran a set of realistic attacks from Kali against the victim machines to validate — and document — the full attack-to-detection story for each technique, mapped against the MITRE ATT&CK framework where applicable.

Exercise 1 — SSH Brute Force (MITRE ATT&CK T1110)

Ubuntu Server's SSH authentication log (/var/log/auth.log) was added as a monitored log source in the Wazuh agent configuration, then Hydra was used from Kali to brute-force SSH logins against Ubuntu Server using a username/password wordlist combination.

```
pope@popesays:~$ ls -la /var/log/auth.log
-rw-r--r-- 1 syslog adm 35672 Jul 13 12:24 /var/log/auth.log
pope@popesays:~$ sudo tail -20 /var/log/auth.log 2>/dev/null
[sudo: authenticat] Password:
2026-07-13T10:31:34.024782+00:00 popesays sudo: pam_unix(sudo:session): sessi
on opened for user root(uid=0) by pope(uid=1000)
2026-07-13T10:31:34.027342+00:00 popesays sudo: pope : TTY=/dev/tty1 ; PWD=/h
ome/pope ; USER=root ; COMMAND=/usr/bin/timedatectl set-timezone Africa/Lagos
2026-07-13T10:31:34.090125+00:00 popesays sudo: pam_unix(sudo:session): sessi
on closed for user root
2026-07-13T10:34:16.035747+00:00 popesays sudo: pam_unix(sudo:session): sessi
on closed for user root
2026-07-13T10:34:24.498545+00:00 popesays sshd-session[3099]: syslogin_perfor
m_logout: logout() returned an error
2026-07-13T10:34:24.507526+00:00 popesays sshd-session[3220]: Received discon
nect from 192.168.1.100 port 44202:11: disconnected by user
2026-07-13T10:34:24.508300+00:00 popesays sshd-session[3220]: Disconnected fr
om user pope 192.168.1.100 port 44202
2026-07-13T10:34:24.510630+00:00 popesays sshd-session[3099]: pam_unix(sshd:s
ession): session closed for user pope
2026-07-13T10:34:24.528503+00:00 popesays systemd-logind[1545]: Session 1 log
ged out. Waiting for processes to exit.
2026-07-13T10:34:24.534541+00:00 popesays systemd-logind[1545]: Removed sessi
on 1.
2026-07-13T10:43:48.254739+00:00 popesays sudo: pam_unix(sudo:session): sessi
on opened for user root(uid=0) by pope(uid=1000)
2026-07-13T10:43:48.255740+00:00 popesays sudo: pope : TTY=/dev/tty1 ; PWD=/h
ome/pope ; USER=root ; COMMAND=/usr/bin/grep -i starting /var/ossec/logs/osse
c.log
2026-07-13T10:43:48.259553+00:00 popesays sudo: pam_unix(sudo:session): sessi
on closed for user root
2026-07-13T11:17:01.370689+00:00 popesays CRON[4070]: pam_unix(cron:session):
session opened for user root(uid=0) by root(uid=0)
2026-07-13T11:17:01.395120+00:00 popesays CRON[4070]: pam_unix(cron:session):
session closed for user root
2026-07-13T11:24:22.157778+00:00 popesays sshd-session[4082]: Accepted passwo
rd for pope from 192.168.1.100 port 39144 ssh2
2026-07-13T11:24:22.159851+00:00 popesays sshd-session[4082]: pam_unix(sshd:s
ession): session opened for user pope(uid=1000) by pope(uid=0)
2026-07-13T11:24:22.167353+00:00 popesays systemd-logind[1545]: New session '
6' of user 'pope' with class 'user' and type 'tty'.
2026-07-13T11:26:24.558667+00:00 popesays sudo: pam_unix(sudo:session): sessi
on opened for user root(uid=0) by pope(uid=1000)
2026-07-13T11:26:24.565516+00:00 popesays sudo: pope : TTY=/dev/pts/0 ; PWD=/
home/pope ; USER=root ; COMMAND=/usr/bin/tail -20 /var/log/auth.log
pope@popesays:~$ sudo nano /var/ossec/etc/ossec.conf
pope@popesays:~$
```

Figure 6.1 — auth.log confirmed active and populated, then explicitly added as a monitored log source in the Wazuh agent's ossec.conf.

The screenshot shows the Wazuh Threat Hunting interface with a table of events and a terminal window in the background.

timestamp	agent.name	rule.description	count	score
Jul 13, 2026 @ 12:54:54.070	UbuntuServer	syslog: User missed the password more than one time	5	5710
Jul 13, 2026 @ 12:54:54.066	UbuntuServer	sshd: Attempt to login using a non-existent user	8	5758
Jul 13, 2026 @ 12:54:54.064	UbuntuServer	Maximum authentication attempts exceeded.	10	2502
Jul 13, 2026 @ 12:54:53.981	UbuntuServer	syslog: User missed the password more than one time	5	5710
Jul 13, 2026 @ 12:54:53.978	UbuntuServer	sshd: Attempt to login using a non-existent user	8	5758
Jul 13, 2026 @ 12:54:53.973	UbuntuServer	Maximum authentication attempts exceeded.	5	5710
Jul 13, 2026 @ 12:54:52.061	UbuntuServer	sshd: Attempt to login using a non-existent user	5	5710
Jul 13, 2026 @ 12:54:51.970	UbuntuServer	sshd: Attempt to login using a non-existent user	5	5710
Jul 13, 2026 @ 12:54:50.083	UbuntuServer	syslog: User missed the password more than one time	10	2502
Jul 13, 2026 @ 12:54:50.075	UbuntuServer	syslog: User missed the password more than one time	10	2502
Jul 13, 2026 @ 12:54:50.073	UbuntuServer	sshd: Attempt to login using a non-existent user	5	5710
Jul 13, 2026 @ 12:54:50.071	UbuntuServer	Maximum authentication attempts exceeded.	8	5758
Jul 13, 2026 @ 12:54:50.069	UbuntuServer	sshd: Attempt to login using a non-existent user	5	5710
Jul 13, 2026 @ 12:54:50.067	UbuntuServer	Maximum authentication attempts exceeded.	8	5758
Jul 13, 2026 @ 12:54:50.064	UbuntuServer	sshd: Attempt to login using a non-existent user	5	5710
Jul 13, 2026 @ 12:54:50.060	UbuntuServer	sshd: Attempt to login using a non-existent user	5	5710
Jul 13, 2026 @ 12:54:49.991	UbuntuServer	syslog: User missed the password more than one time	10	2502
Jul 13, 2026 @ 12:54:49.985	UbuntuServer	sshd: Attempt to login using a non-existent user	5	5710

The terminal window in the background shows a Hydra brute-force attack in progress against the SSH service on 192.168.1.103. It displays the command used, the wordlists, and the progress of the attack, including the number of login attempts and the current status.

Figure 6.2 — Hydra brute-force attack in progress against Ubuntu Server's SSH service, alongside the Wazuh Threat Hunting view showing real-time detections escalating from individual failed-login alerts to correlation-rule alerts ("Maximum authentication attempts exceeded").

This exercise demonstrated genuine SIEM correlation behavior — Wazuh did not just log individual failed logins, it recognized the repeating pattern from a single source and escalated the alert severity accordingly, in addition to Snort independently flagging the same activity at the network layer.

Exercise 2 — Vulnerability Scanning & Exploit Research (MITRE ATT&CK T1595 / T1190)

Nmap's vulnerability scanning script engine was used to fingerprint Ubuntu Server's exposed SSH service and cross-reference its version against known CVEs.

```

[~] (paper@paper) ~$ nmap -sV --script vuln 192.168.1.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-13 13:16 +0100
Nmap scan report for 192.168.1.102
Host is up (0.00023s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 10.2p1 Ubuntu 2ubuntu3.2 (Ubuntu Linux; protocol 2.0)
|_vulners:
|_cpe:/a:openssh:openssh:10.2p1:
|_CVE-2026-68002 9.4 https://vulners.com/cve/CVE-2026-68002
|_CVE-2026-35414 6.1 https://vulners.com/cve/CVE-2026-35414
|_CVE-2026-35386 6.1 https://vulners.com/cve/CVE-2026-35386
|_CVE-2026-35385 6.1 https://vulners.com/cve/CVE-2026-35385
|_CVE-2026-60000 7.5 https://vulners.com/cve/CVE-2026-60000
|_CVE-2026-59999 7.5 https://vulners.com/cve/CVE-2026-59999
|_CVE-2026-60001 6.5 https://vulners.com/cve/CVE-2026-60001
|_CVE-2026-59998 6.5 https://vulners.com/cve/CVE-2026-59998
|_CVE-2026-35387 6.5 https://vulners.com/cve/CVE-2026-35387
|_CVE-2026-59997 5.4 https://vulners.com/cve/CVE-2026-59997
|_CVE-2026-59996 5.4 https://vulners.com/cve/CVE-2026-59996
|_CVE-2026-59995 5.4 https://vulners.com/cve/CVE-2026-59995
|_CVE-2026-35388 7.5 https://vulners.com/cve/CVE-2026-35388
MAC Address: 08:0C:29:1E:58:AA (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/report/
Nmap done: 1 IP address (1 host up) scanned in 14.08 seconds

```

Figure 6.3 — Nmap vulnerability scan identifying OpenSSH and listing associated CVEs with CVSS scores, several rated High/Critical.

To move from "a CVE exists" to "a CVE is practically actionable," Exploit-DB was searched locally via searchsploit for any working exploit matching the installed OpenSSH version.

Exploit Title	Path
Debian OpenSSH - (Authenticated) Remote SEL	linux/remote/6094.txt
Dropbear / OpenSSH Server - 'MAX_UNAUTH_CLI	multiple/dos/1572.pl
FreeBSD OpenSSH 3.5p1 - Remote Command Exec	freebsd/remote/17462.txt
glibc-2.2 / openssh-2.3.0p1 / glibc 2.1.9x	linux/local/258.sh
Novell Netware 6.5 - OpenSSH Remote Stack O	novell/dos/14866.txt
OpenSSH 1.2 - '.scp' File Create/Overwrite	linux/remote/20253.sh
OpenSSH 2.3 < 7.7 - Username Enumeration	linux/remote/45233.py
OpenSSH 2.3 < 7.7 - Username Enumeration (P	linux/remote/45210.py
OpenSSH 2.x/3.0.1/3.0.2 - Channel Code Off-	unix/remote/21314.txt
OpenSSH 2.x/3.x - Kerberos 4 TGT/AFS Token	linux/remote/21402.txt
OpenSSH 3.x - Challenge-Response Buffer Ove	unix/remote/21578.txt
OpenSSH 3.x - Challenge-Response Buffer Ove	unix/remote/21579.txt
OpenSSH 4.3 p1 - Duplicated Block Remote De	multiple/dos/2444.sh
OpenSSH 6.8 < 6.9 - 'PTY' Local Privilege E	linux/local/41173.c
OpenSSH 7.2 - Denial of Service	linux/dos/40888.py
OpenSSH 7.2p1 - (Authenticated) xauth Comma	multiple/remote/39569.py
OpenSSH 7.2p2 - Username Enumeration	linux/remote/40136.py
OpenSSH < 6.6 SFTP (x64) - Command Executio	linux_x86-64/remote/45000.c
OpenSSH < 6.6 SFTP - Command Execution	linux/remote/45001.py
OpenSSH < 7.4 - 'UsePrivilegeSeparation Dis	linux/local/40962.txt
OpenSSH < 7.4 - agent Protocol Arbitrary Li	linux/remote/40963.txt
OpenSSH < 7.7 - User Enumeration (2)	linux/remote/45939.py
OpenSSH SCP Client - Write Arbitrary Files	multiple/remote/46516.py
OpenSSH server (sshd) 9.8p1 - Race Conditio	linux/remote/52269.c
OpenSSH/PAM 3.6.1p1 - 'gossh.sh' Remote Use	linux/remote/26.sh
OpenSSH/PAM 3.6.1p1 - Remote Users Discover	linux/remote/25.c
OpenSSHd 7.2p2 - Username Enumeration	linux/remote/40113.txt
Portable OpenSSH 3.6.1p-PAM/4.1-SuSE - Timi	multiple/remote/3303.sh
Shellcodes: No Results	

Figure 6.4 — searchsploit results: every publicly available exploit module targets older OpenSSH versions — none apply to the installed version.

Finding: although Nmap's version-matching flagged several CVEs (including one scored 9.4/10), no publicly available exploit code targets the installed OpenSSH version. This is documented as a deliberate analyst conclusion: a scan finding is not automatically an actionable exploit path, and vulnerability triage requires checking exploit availability separately from CVE existence. Patching remains advisable regardless, since exploit code could be developed or disclosed later.

Investigative Use of Wazuh — Attribution

Beyond passive alerting, Wazuh's Threat Hunting search was used analyst-style to pivot on the attacker's IP address across all monitored agents, reconstructing which machines Kali had touched and in what order — the same technique a SOC analyst would use to scope the extent of an intrusion once an indicator is known.

7. Troubleshooting Log (Consolidated)

The table below consolidates every significant issue encountered across all phases, in the order they occurred, as a single reference — useful both as a rebuild checklist and as evidence of methodical problem-solving.

#	Phase	Issue	Resolution
1	A	pfSense installer failed ("system is NOT installed")	Increased VM RAM 256MB → 2048MB; confirmed WAN/NAT internet access for package fetch
2	B	Two DHCP servers on VMnet2 causing lease conflicts	Disabled VMware's local DHCP service on VMnet2, leaving pfSense as sole DHCP authority
3	B	apt update failing with DNS resolution errors	Verified/restarted pfSense's DNS Resolver service; confirmed DHCP-advertised DNS
4	C	Custom Snort rule rejected: "Illegal direction specifier"	Re-typed rule manually to eliminate a corrupted arrow character from copy-paste
5	C	Snort produced zero alerts despite live cross-host traffic	Diagnosed as VMware blocking promiscuous mode; fixed via .vmx flags + device permissions + full VM power-cycle
6	D	Wazuh agent .deb download returned 403 Forbidden	Used the Wazuh dashboard's auto-generated Deploy Agent command instead of a manual URL
7	D	Ubuntu agent enrollment failing (wrong target IP)	Corrected the manager address in ossec.conf (was pointing at itself, not the manager)
8	D	Sysmon install: "Sysmon64.exe is not recognized"	Navigated to the correct extracted-file directory before running the install command
9	D	pfSense remote syslog form rejected save	Combined IP:port into a single field instead of two separate boxes
10	D	Sudden burst of rootcheck alerts looked like a loop	Confirmed via ossec.log timestamps this was expected scan-on-start behavior tied to known restarts
11	D	Snort alerts not appearing in Wazuh dashboard	Identified as an architectural gap — pfSense's Snort package does not use the general syslog forwarder; documented as a known limitation

8. Lessons Learned

Key Takeaways

- Network segmentation and DHCP authority must be unambiguous — running two DHCP servers on one segment produces confusing, intermittent symptoms that look like unrelated bugs elsewhere in the stack.
- Hypervisor-level security controls (like VMware's promiscuous-mode block) can silently defeat a correctly-configured IDS. Verifying with an independent tool (tcpdump) rather than trusting the guest OS's own view of its interface flags was the key diagnostic step.
- A SIEM's value comes from correlation, not just log collection — the brute-force exercise showed Wazuh escalating repeated failures into higher-severity alerts automatically, which is what actually makes an alert queue usable at scale.
- Not every integration path is native — pfSense's Snort package not forwarding through general syslog is a real architectural boundary, and documenting a limitation honestly is more valuable than forcing a fragile workaround.
- A vulnerability scan finding a CVE is not the same as a viable attack — checking exploit availability before assuming impact is a basic but important vulnerability-management discipline.
- Before assuming a fault, verify against evidence — the rootcheck alert burst looked alarming but was resolved by simply reading timestamps in the log rather than reflexively changing configuration.

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